

Pairing Proofs and Polynomial Ring Toolkit

Shramee Srivastav
MIST.cash—FOCBB, Shhtarknet
Email: shramee@proton.me

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Abstract

We present a two-round public-coin interactive proof system for verifying optimal ate pairings on Barreto–Naehrig curves, together with its Fiat–Shamir-compiled non-interactive variant. Our central technique is to express each step of the pairing computation—a Miller loop doubling, a doubling-and-addition, a Frobenius correction, or the residue-witness final exponentiation check of Novakovic and Eagen (CRYPTO 2024)—as a *single* multi-operand polynomial identity over $\mathbb{F}_q[x]$ modulo the irreducible P_{12} defining $\mathbb{F}_{q^{12}}$, verified probabilistically via the Schwartz–Zippel lemma. The quotients arising from all such identities across an entire pairing are then collapsed into a single accumulated polynomial via a random linear combination, reducing the verifier’s work to one Euclidean division check at a single random evaluation point.

Compared to prior polynomial-identity approaches that operate at the granularity of individual $\mathbb{F}_{q^{12}}$ multiplications (Feltroidprime, 2023), our coarser granularity eliminates intermediate remainder commitments and unifies the Miller loop and final exponentiation under one verification framework. We prove knowledge soundness $\delta_s \leq (n - 1 + d_x)/|\mathbb{F}_q|$ in the interactive setting, yielding $\delta_s < 2^{-244}$ for BN254 after the Fiat–Shamir transform in the random oracle model.

The construction is motivated by, and directly applicable to, arithmetized verification of pairings inside SNARK circuits (R1CS, Plonkish, AIR), where each $\mathbb{F}_q$ multiplication and each prover commitment carries a significant cost. As a reusable artefact we contribute a polynomial-ring toolkit for the **emulated** package of **gnark**, parameterised by an arbitrary emulated field and modulus polynomial. Benchmarked on a Groth16 verifier circuit over BN254, the toolkit yields a 39% reduction in Plonkish constraints (1,890,771 $\rightarrow$ 1,158,194), a 60% reduction in Fiat–Shamir commitments (739,198 $\rightarrow$ 295,946), and up to a 52% reduction in solver memory over the upstream **gnark** baseline. The same technique is already deployed in the Garaga library on Starknet.

1 Introduction

Blockchain technology continues to mature with improving scalability and infrastructure. However, privacy remains a significant barrier to entry for many users and applications. Zero-knowledge proofs (ZKPs) have emerged as a powerful solution to this challenge.

1.1 Motivation

Elliptic curve pairings represent fundamental building blocks for numerous cryptographic systems including Groth16 [Gro16], PLONK [GWC19], BLS signature schemes [BGW⁺22], and KZG polynomial commitment schemes [KZG10].

However, **pairings are computationally intensive**. This creates critical bottlenecks in two important deployment contexts:

1. **Resource-constrained virtual machines** (Ethereum VM, BitVM, ZKVMs): Computation is metered — each operation incurs gas costs. A single pairing verification requiring thousands of field operations becomes economically prohibitive during peak network congestion.
2. **Arithmetized computation systems** (R1CS [GGPR13], Plonkish [GWC19], Cairo AIR [GPR21]): Each operation must be represented algebraically, adding significant prover overhead beyond native execution costs.

In such contexts, an efficient proof system for pairing correctness can replace expensive native execution inside the constrained environment.

1.2 Our Contribution

We present a public-coin interactive proof system for verifying pairing computations that consolidates verification into unified polynomial relationships under the Schwartz-Zippel lemma. By treating complete Miller loop iterations as single polynomial relationships rather than sequences of individual extension field operations [Fel23], our approach achieves:

- **39% reduction** in SCS constraints ($1,890,771 \rightarrow 1,158,194$ for BN254 Groth16 simulation)
- **60% reduction** in commitments ($739,198 \rightarrow 295,946$)
- **~42% reduction** in compile-time memory and up to **52%** in solver memory

Our key insight is that instead of verifying individual field operations separately, we can verify entire Miller loop iterations as single polynomial identities. This consolidation dramatically reduces both the number of modular operations required and the communication overhead from intermediate commitments.

As a reusable artefact of independent interest, we also contribute a **polynomial ring toolkit** for gnark’s emulated arithmetic package. The toolkit packages the two-round interactive proof system into a clean API: a single `NewPolyRingCheck` call registers a modulus, `MulPolyRings` submits multiplications with deferred verification, and `performDeferredRingChecks` closes the circuit with two `api.Commit` challenges. Atop this, the `PolyRingAccumulator` provides a higher-level interface for building product chains: factors are enqueued via `Mul`, squarings via `Sqr`, and the accumulated product is evaluated lazily through `Eval`, with optional automatic degree-bounding to keep intermediate products manageable.

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